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Mercury supply limits methylmercury production in paddy soils

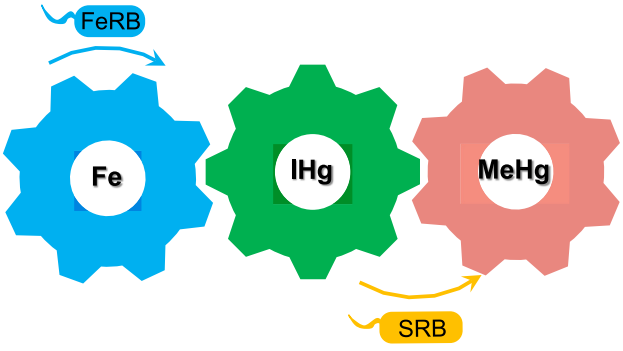
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HIGHLIGHTS

- Hg supply instead of SRB activities limits Hg methylation in paddies.
- SRB are the primary Hg methylators in non-Hg contaminated paddies.
- FeRB mediate Hg methylation via regulating Hg mobility.
- The mobilization of inert Hg drives Hg methylation under changing environments.

GRAPHICAL ABSTRACT



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ABSTRACT

The neurotoxic methylmercury (MeHg) is a product of inorganic mercury (IHg) after microbial transformation. Yet it remains unclear whether microbial activity or IHg supply dominates Hg methylation in paddies, hotspots of MeHg formation. Here, we quantified the response of MeHg production to changes in microbial activity and Hg supply using 63 paddy soils under the common scenario of straw amendment, a globally prevalent agricultural practice. We demonstrate that the IHg supply is the limiting factor for Hg methylation in paddies. This is because IHg supply is generally low in soils and can largely be facilitated (by 336–747 %) by straw amendment. The generally high activities of sulfate-reducing bacteria (SRB) do not limit Hg methylation, even though SRB have been validated as the predominant microbial Hg methylators in paddies in this study. These findings caution against the mobilization of legacy Hg triggered by human activities and climate change, resulting in increased MeHg production and the subsequent flux of this potent neurotoxin to our dining tables.

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1. Introduction

Paddy soils are a kind of artificial wetland and have been identified as hotspots of Hg methylation (Tang et al., 2020). As wetlands, rice paddies are distributed in 117 countries and cover an area of over 140 million ha globally (Global Rice Science Partnership, 2013), playing an indispensable role in food production and provision of ecosystem services. For instance, rice paddies serve as habitats for 624 pest species and >100 common weed species (Luo et al., 2014). As a result of anthropogenic management, paddy soils receive massive input of organic matter (OM), nutrient elements like nitrogen (N), phosphorous (P), and potassium (K), and hydrological alterations (Global Rice Science Partnership, 2013; Kögel-Knabner et al., 2010). This results in the enrichment of Hg from deposition, irrigation, and fertilization (Liu et al., 2023; Soares et al., 2015; Xue et al., 2019) and enhances the biomass of anaerobic microbes (Wei et al., 2022), making paddy soil an ideal place to produce neurotoxic MeHg. The massive production of MeHg in paddy soils leads to risks of MeHg exposure to humans via rice consumption (Gong et al., 2018) and to inhabited wildlife (Windham-Myers et al., 2014). Additionally, the rice-paddy systems could also be a significant source of MeHg for downstream lakes and rivers through runoff (Brigham et al., 2009), contributing to MeHg accumulation in aquatic food chains. Therefore, there is increasing interest in exploring the biogeochemical cycling of Hg in paddy soils and the associated risks.

In paddy soils, MeHg is primarily derived from the microbial transformation of inorganic Hg (IHg) (Hsu-Kim et al., 2013), but it remains unclear whether microbial activity or IHg supply is the predominant factor limiting MeHg production. Previous studies have paid much attention to microbial Hg methylators, for instance, revealing the composition and the roles of microbial methylators (Liu et al., 2018; Ma et al., 2019; Pu et al., 2022) and the impacting factors, such as the pH, organic matter content, and C/N ratio of soils (Liu et al., 2019b). Enhanced microbial activities were reported to be primarily responsible for the elevated MeHg under a number of scenarios, such as OM input (Tang et al., 2019a; Zhao et al., 2018), water management (Wang et al., 2014), and biochar application (Zhang et al., 2018). By contrast, less attention has been paid to IHg supply, among which more attention has been paid to IHg immobilization by Se (Wang et al., 2016; Zhang et al., 2015; Zhang et al., 2014), minerals (Li et al., 2014; Xiang et al., 2022), OM (Sun et al., 2022a; Wang et al., 2020), etc., but not IHg mobilization. Recently, the existence of nano-HgS (Lei et al., 2022) and the mobilization of HgS after sulfate addition (Li et al., 2022b; Li et al., 2019b) has been reported, as well as their nonnegligible contributions to MeHg production (Li et al., 2022a; Liu et al., 2019a; Tian et al., 2021). These pioneering studies suggest the potential role of IHg supply in controlling MeHg formation in paddy soils, yet the importance compared to microbial activity requires further evaluation. Due to the knowledge gap of the limiting factor for Hg methylation in paddy soils, it is challenging to efficiently mitigate MeHg exposure via rice consumption.

Additionally, most studies on Hg methylation in paddy soils were conducted using Hg-contaminated or Hg mining-impacted soils, with limited implications on uncontaminated paddy soils. This is because Hg species are quite different between the uncontaminated (primarily OM-bound (Liang et al., 2019)) and the Hg-mining impacted soils (mostly inert HgS (Tang et al., 2019a)), especially considering that different Hg species have distinct availability to microbial methylators (Jonsson et al., 2012). These differences in Hg species and microbial community (Pu et al., 2022) make it challenging to extrapolate the mechanisms from previous studies to uncontaminated soils, where Hg methylation potentials are generally higher than those in contaminated/mining-impacted soils (Tang et al., 2020). Furthermore, Hg in rice grains has been seen to exceed the national standard set for total Hg (THg) (i.e., 20 μg/kg in China) even when the soil THg concentrations are below the screen value (0.5 mg/kg) (Hang et al., 2018). This suggests the necessity to study Hg methylation in non-contaminated paddies. Identifying the limiting factor for MeHg production in paddy soils requires quantifying changes in MeHg production in soils varied in soil properties, as well as quantifying the response of MeHg production to changes in microbial activities and Hg supply in changing environments.

Here, we used 63 uncontaminated paddy soils to identify the limiting factor for Hg methylation in paddy soils (Fig. S1). Straw amendment in paddy soils is widely adopted as a way to increase soil fertility and has been reported to increase Hg mobility, microbial activities, and the subsequent net production of MeHg (Sun et al., 2022a; Tang et al., 2018; Zhu et al., 2016). Briefly, 3 batch incubation experiments were designed to explore 1) the spatial variations in MeHg production, 2) temporal variations in Hg methylation, and 3) the relative contributions of primary microbial methylators, respectively. By analyzing the correlations between MeHg production and the associated processes/parameters using methods such as machine learning, the limiting factor for Hg methylation in paddy soils was identified. Results obtained from this study can advance the understanding of Hg methylation and provide insights into more effective Hg risk mitigation.

2. Material and methods

2.1. Experimental design

Paddy soils and rice straw. A total of 63 types of surface paddy soils were used in this study. These soils were collected from 12 provinces, and the total rice planting areas of those provinces together account for 83 % of that in China (China Statistical Yearbook 2021). The geographic information of these soils is depicted in Fig. S1 (Supplementary Information, SI). Soils from different sites were collected at the same depth of 1–13 cm because this is a puddling soil layer recognized as a hot spot of Hg methylation (Zhao et al., 2016). After being collected in the fields, soils were immediately brought to the laboratory in ice boxes, ground, sieved through a 2-mm mesh, and stored in the refrigerator (4 °C) before parameter analysis (e.g., water content, total Hg and MeHg concentrations, etc.) and experimentation.

The rice straw used in this study was obtained from a paddy field in Yixing, Jiangsu Province, China, and was free of any known Hg contamination. Rice straw was rinsed thoroughly with deionized water, over-dried, and then crushed into powder, which was sieved through a 150-μm mesh before incubation. This is to mimic real-world agricultural practices where straw is commonly chopped into small pieces to enhance its decomposition in situ. The concentrations of total Hg (THg) and MeHg in the rice straw were determined to be 30.9 ± 0.2 and 2.7 ± 0.1 μg/kg, respectively.

Incubation experiment I-Spatial variations. To quantify the response of MeHg production on a large scale, 63 soils were subjected to an incubation experiment with or without the amendment of rice straw. For each soil, two treatments, i.e., CK (soil without rice straw) and RS (soil with 1 % rice straw amendment), were established, and each treatment contained three replicates. Specifically, 5 g of soil (dry weight) with or without straw powder was put into a 50-mL polypropylene centrifuge tube (Corning, USA). Then 30 mL of deionized water was added to each tube. After thorough mixing, the tubes were sealed and subjected to a 21-d incubation in an artificial climate chamber (28 °C in the dark). An incubation duration of 21 days was selected mainly because soil MeHg concentrations could reach the peaks within 21 days according to our previous study (Tang et al., 2019a) and Sun et al. (2022a), under which MeHg production is sensitive to key parameters and thus the limiting factors could be identified.

At the end of the incubation, all samples were centrifuged at 4000 rpm for 20 min and then uncapped in an anaerobic glove box to separate the overlying water and the soil. The overlying water (simulating pore water in paddy soils) was measured for pH and Eh (HACH, HQ30D, USA) and filtered through a 0.45-μm polyether sulfone filter capsule (Anpel, China) for the determination of dissolved organic carbon/DOC (Elementar vario TOC cube, Element, Germany) and sulfate concentrations (Ion chromatography, Dionex, ICS-1000, USA). The soil particles were

subjected to the analyses of water content, MeHg concentrations, HCl-extractable Fe(II) (indicative of the activity of iron-reducing bacteria/FeRB, details below), *hgcA* gene copies, and constants of Hg methylation rate (k_m) and MeHg demethylation rate (k_d).

Incubation experiment II-Temporal variations. Eight soils (i.e., CZ, HK, HY, HZ, NC, MDJ, TR, and YA) were used to further elucidate the potential mechanisms of Hg methylation in paddy soils. These 8 soils were selected due to the wide geological distribution and large variations in constant rates of methylation and demethylation according to the results obtained from the national survey. Two treatments, CK (soil without rice straw) and RS (soil +1 % rice straw amendment), were set as described above. After incubation for 7, 14, 21, and 35 days, three tubes from each treatment were randomly selected and centrifuged to separate soil particles from the overlying water. The overlying water was filtered through a 0.45-μm polyether sulfone filter capsule for the analysis of dissolved THg (Brooksrand, USA) and the spectral characteristics of dissolved organic matter (DOM) (for samples collected on days 0, 7, 14, and 21; HITACHI F-7000, Japan). The DOM components, including aromatic protein I, aromatic protein II, fulvic acid-like, microbial byproduct-like, and humic acid-like fraction, were quantified by the fluorescence region integration method (Chen et al., 2003). The wet soil particles were used for the quantification of HCl-extractable Fe(II) and MeHg concentrations. The soil samples collected on day 21 were also subjected to the analysis of the composition of the microbial community using 338F/806R as the primers (Klindworth et al., 2013) (Shanghai Majorbio Bio-pharm Technology Co., Ltd).

Incubation experiment III-Metabolism inhibition assays. Metabolism inhibition assays were conducted to identify the contributions of major *hgcA*-containing microorganisms to MeHg production. Sulfate-reducing bacteria (SRB), iron-reducing bacteria (FeRB), and methanogens are the most important methylation microorganisms (Parks et al., 2013). The same 8 soils used in the Incubation experiment II were selected. For each soil, 4 treatments (3 replicates for each treatment) were set: (1) soils without adding any inhibitors or straw (CK), (2) soils added with 1 % straw (RS), (3) soils added with 1 % straw and inhibitor of SRB activities (20 mM molybdate, RS-SRB) (Wu et al., 2020), and (4) soils added with 1 % straw and inhibitor of methanogens activities (30 mM bromoethanesulfonate, RS-Meth) (Tang et al., 2019a).

Specifically, 5 g of soil (dry weight) was re-suspended in 30 mL deionized water in a 50 mL centrifuge tube, with the additions of specific inhibitors described above. The caps of the centrifuge tubes were pre-punched, and then the hole was sealed with silicone rubber, from which air was sampled using an injection syringe for CH_4 determination (gas chromatography, 7890B, Agilent, details can be found in Lei et al. (2021)). After 21 days of incubation, all tubes were used to measure the methane production in the air (indicating methanogen activities) first and then centrifuged (4000 rpm, 20 min) to separate the soil and the overlying water. The MeHg concentrations and HCl-extractable Fe(II) in soils were determined, and the overlying water was measured for sulfate.

2.2. Parameter analyses

Soil MeHg and dissolved THg. The soil MeHg was extracted using KOH-CH_3OH solution based on the method recommended by the Florida Department of Environmental Protection (Bloom, 1989; Xu et al., 2020) and was analyzed based on EPA Method 1630. Our previous study comfirmed the high extraction efficiency and the negligible production of MeHg artifacts for this extracting approach (Tang et al., 2023). Briefly, ~0.5–0.8 g (wet weight) of soil was put into a 15 mL centrifuge tube, and then 2 mL of 25 % (*w*/w) KOH-CH_3OH was added. Samples were subjected to MeHg extraction in a shaker (230 rpm) at 70 °C for 4 h in the dark. And then, the digested solutions were used to determine soil MeHg concentration using an automatic MeHg analyzer (Brooks Rand, Brooks Rand model III). The detection limit was 0.05 pg, and the recovery of the standard reference material (ERM-CC580, MeHg 75 ± 4 μg/kg) was 87 %–104 %. The concentration of THg in the overlying water was analyzed according to EPA Method 1631 using an automatic Hg analyzer (Brooks Rand model III, Brooks Rand) after the digestion using BrCl.

k_m and k_d. The constants of Hg methylation and MeHg demethylation rates were quantified using isotopic tracer techniques based on the method established by Hintelmann et al. (2000). Specifically, 5 g of soil and 30 mL of deionized water were put into a 50 mL centrifuge tube and incubated as described in incubation experiment I. On each sampling day, 20–50 μL of ^{202}Hg and $Me^{199}Hg$ were added at concentrations of 10–80 % of ambient THg and MeHg levels, and then the soils were mixed and incubated for 24 h. After incubation, soils were extracted as described above for isotopic MeHg analysis. The concentrations of $Me^{202}Hg$ (used to calculate methylation rate, k_m), $Me^{201}Hg$ (as internal standard), $Me^{200}Hg$ (used to calculate ambient MeHg), and $Me^{199}Hg$ (used to calculate demethylation rate, k_d) were determined by GC-ICP-MS (Perkin Elmer, NexION300). Standard reference material (ERM-CC580) was included in each isotopic analytical batch, and the recovery rates based on the isotopic calculation were 86.5 % ± 6.7 (n = 15, 77.6–98.1 %). The constant rates of Hg methylation and MeHg demethylation were calculated as follows:

$$k_m = \frac{\left[Me^{202}Hg^{+}\right]_{t_x}}{\left[^{202}Hg^{2+}\right]_{t0} \cdot t} \quad (1)$$

$$k_d = \frac{ln\left[Me^{199}Hg^{+}\right]_{t_0} - ln\left[Me^{199}Hg^{+}\right]_{t_x}}{t} \quad (2)$$

Where *[MeHg$^+$]* and *[Hg$^+$]* represent the concentration of MeHg and IHg; t_0 refers to the start time, t_x is referred to as the end time, and t is the incubation time (24 h).

***HgcA* gene quantification.** DNA was extracted from the freeze-dried soils according to the manual of MoBio PowerSoil® DNA Isolation Kit (QIAGEN Inc., USA) (Liu et al., 2018). The concentration and quality of the isolated DNA were determined by a spectrophotometer (NanoDrop ND2000c UV–vis, Thermo Fisher Scientific, USA). Then the qualified DNA samples were diluted 10-fold for real-time quantitative PCR. Two pairs of primers were prepared for *Deltaproteobacteria* and *Archaea*, respectively: the ORNL-Delta-HgcA-F (5'-GCCAACTACAAGMTGASCWC-3′) and ORNL-Delta-HgcA-R (5'-CCSGCNGCRCACCAGACRTT-3′) for *Deltaproteobacteria* group with *hgcA* gene, while the ORNL-Archaea-HgcA-F (5'-AAYTAYWCNCTSAGYTTYGAYGC-3′) and ORNL-Archaea-HgcA-R (5'-TCDGTCCCRAABGTSCCYTT-3′) for *Archaea* group with *hgcA* gene (Christensen et al., 2016). The quantitative assay system of *hgcA* copy number contained 25 μL solution, including 12.5 μL of SYBR Premix *Ex Taq* (Takara Bio Inc., Japan), 0.5 μL of *Deltaproteobacteria* or *Archaea* forward and reverse primer, 9.5 μL sterilized water, and 2 μL of diluted DNA. The thermal-cycling conditions were 3 min of initial denaturation at 95 °C, 40 cycles of 15 s at 95 °C, 15 s at 50 °C, 15 s at 55 °C, and 4 min at 72 °C, followed by a plate read at 83 °C for *Deltaproteobacteria*; and were 3 min of initial denaturation at 95 °C, 40 cycles of 15 s at 95 °C, 30 s at 50 °C, and 25 s at 55 °C, then 4 min at 72 °C, followed by a plate read at 83 °C for *Archaea* (Bravo et al., 2018).

HCl-extractable Fe(II). The content of HCl-extractable Fe(II) was quantified based on the 1,10-phenanthroline colorimetric method (Julian et al., 2017; Tong et al., 2014). Briefly, ~0.4 g wet soil was weighted into a 15-mL centrifuge tube and extracted with 3 mL of 0.5 M HCl for 1 h. The extracting solution was then centrifuged at 4000 r/min for 10 min to obtain the supernatant. Then, 19.3 mL of deionized water, 0.5 mL of 0.1 % 1,10-phenanthroline, and 200 μL supernatant were added to a 50-mL centrifuge tube. After adjusting the solution pH to 4–5 using 10 % (*w*/w) sodium acetate, the absorbency at 510 nm was measured using a spectrophotometer (HACH, DR 1900).

2.3. Statistical analyses and explainable machine learning

Differences between treatments were determined using a one-way analysis of variance (ANOVA, SPSS, version 16.0). Posthoc tests were used to identify variances among groups (Tukey, $p < 0.05$).

Multiple linear regression (stepwise) and 13 other machine learning algorithms were applied to identify the key parameters impacting soil MeHg. A total of 126 groups of data sets obtained from the incubation experiment I, including 63 CK groups and 63 RS groups, was used for this purpose. Such a sample size is acceptable for machine learning (Palansooriya et al., 2022; Yu et al., 2023). Data obtained from Incubation experiments II and III were not used for machine learning, mainly because some key parameters related to Hg methylation, for example, pH and Eh, were not monitored in these two experiments. For multiple linear regression, the collinearity test was carried out at first, and the variance inflation factors (VIF) were < 2; The maximum value of Cook distance and Durbin-Watson value were < 1 and ~ 2, respectively. The remaining thirteen machine learning algorithms included decision trees, bagged decision trees, Bayesian additive regression trees (BART), random forests, XGBoost, Cubist, ridge regression, multivariate adaptive regression splines (MARS), bagged MARS, K-nearest neighbors, and three support vector machines (linear, polynomial, and radial basis function). The entire dataset was divided into the training set (70 %) and the testing set (30 %). A ten-fold cross-validation and a dense grid search were applied to tune the hyperparameters using the training set. Model evaluation was based on root mean square errors (RMSE). The well-reained regression model was interpreted using explainable AI (XAI) methods for both global explanation (feature importance and partial dependency) and local explanation (Shapley Additive exPlanations/ SHAP and the local stability plot).

3. Results

3.1. Spatial variations in MeHg production

The MeHg concentration in paddy soils without straw amendment ranged from 0.44 to 0.92 μg/kg on a provincial scale and increased to 0.93–2.57 μg/kg after straw return (Fig. 1a). The average constants of Hg methylation rate (k_m, %/d) and demethylation rate (k_d, /d) elevated from 0.21 to 0.38 and from 0.64 to 0.99 on the national scale ($p < 0.05$), respectively, with increments of 21–178 % and 0–142 % on the provincial scale (Fig. 1b and c). Surprisingly, the copies of methylation gene *hgcA* remained constant after straw amendment in *Deltaproteobacteria* (8.3×10^8 copies/g soil in CK versus 8.36×10^8 copies/g soil in RS, $p > 0.05$), while decreased slightly but significantly in *Archaea* on the national scale (9.58×10^9 in CK versus 9.36×10^9 in RS, $p < 0.001$, Fig. S2). Straw amendment also showed significant impacts on soil properties, including elevating the pH, DOC, and HCl-extractable Fe(II) concentrations, as well as reducing redox potential and sulfate concentrations in the overlying water (Fig. S3, a–e).

Regardless of the straw amendment, soil MeHg concentrations were significantly correlated to k_m, sulfate, HCl-extractable Fe(II), and DOC concentrations ($p < 0.01$, Fig. S4, a–d) but not to sulfate reduction or the sulfate reduction rate (Fig. S4, e–f). This suggests that the enhanced sulfate reduction is not responsible for the elevated MeHg production, and thus, SRB activity is not the limiting factor for MeHg production in paddy soils, irrespective of straw addition. Interestingly, when utilizing machine learning to predict MeHg concentration, the most important feature variable indicated by feature importance analysis was HCl-extractable Fe(II), followed by Eh, pH, and sulfate (Fig. 2, details in Supporting Text 1 and Fig. S5). Meanwhile, results of the multiple linear regression also showed that HCl-extractable Fe(II), soil pH, and DOC concentrations were the top three key feature variables impacting MeHg concentrations in soils (Table 1 and Fig. S6). Results of multiple linear regression and the XGBoost model suggest the importance of Fe reduction to net MeHg production.

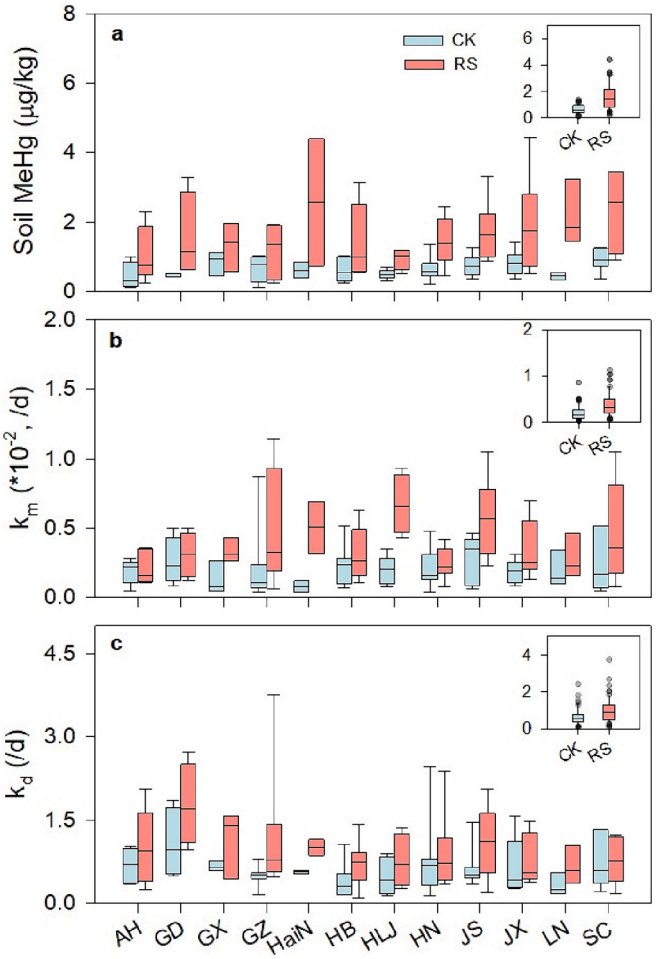


Fig. 1. Spatial variation experiments: MeHg concentrations (a), Hg methylation rates (k_m, b), and MeHg demethylation rates (k_d, c) after 21 days of incubation with or without rice straw addition. CK and RS represent the treatments of control and rice straw amendment, respectively. In each panel, the upper and lower whiskers of the box represent the 95th and 5th percentiles, and the horizontal lines represent the median values. The smaller figures in panels a–c depict the distributions of these parameters on a national scale, with dots representing the outliers.

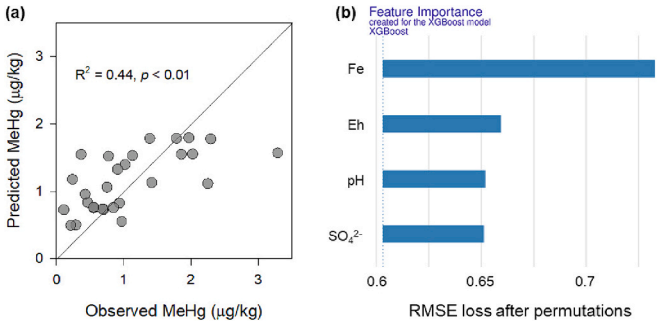


Fig. 2. Performance of XGBoost model and feature importance ranking in predicting soil MeHg concentration. Fe and SO_4^{2-} indicate the concentrations of HCl-extractable Fe(II) in soils (g/kg) and sulfate concentrations in the overlying water (mg/L, $\log_{10}$-transformed). Machine learning algorithm comparisons of 13 models and the explanations of the XGBoost model are presented in Fig. S5 and discussed in Supporting Text 1 (SI).

Table 1
Results of multiple linear regressions for 63 types of paddy soils.

Equations	Adjusted r^2	Significant level
[MeHg] = 0.406 × [Fe] + 0.623	0.19	<0.001
[MeHg] = 0.509 × [Fe] – 0.346 × pH + 2.639	0.23	<0.001
[MeHg] = 0.406 × [Fe] – 0.398 × pH + 0.025 × [DOC] + 2.774	0.26	<0.001

Note: A total of 126 groups of data were used for multiple linear regression. Here, DOC ([DOC], mg/kg), HCl-extractable Fe(II) ([Fe], g/kg), sulfate concentration (mg/L), sulfate reduction (mg/L), sulfate reduction rate (%), k_m (%), k_d, *hgcA*-bacteria (× 10^8 copies/g soil), *hgcA-Archaea* (× 10^9 copies/g soil), pH, and Eh values (V), were taken as independent variables. The maximum value of Cook distance and Durbin-Watson value is 0.536 (< 1) and 1.82 (~2), respectively. The normal distribution histogram and P—P plot are presented in Fig. S6.

3.2. Temporal variations in Hg methylation

To further identify the controlling factor for MeHg production in paddy soils, the dynamics of key parameters related to Hg methylation, including k_m & k_d, dissolved THg, Fe reduction, and DOM composition, were monitored in 8 soils with or without straw amendment. Similarly, soil MeHg, k_m, and k_d increased significantly in most soils after straw amendment. For these 8 tested soils, MeHg increased by 13.5–189 %, 18.4–113.1 %, 2.7–163 %, and 20.1–709.5 % on Days 7, 14, 21, and 35, respectively (Fig. S7). For soil MeHg in both treatments, there was an increasing trend from day 7 to day 21, while it tended to decrease afterward, indicating net Hg methylation at early stages while net demethylation at the later stage of incubation (Fig. S7 and Supporting Text 2). The k_m and k_d increased by 106–174 % and 28–326 % during the 35-day incubation period (Figs. S8–9). Importantly, the concentration of dissolved THg elevated by 747 %, 336 %, 614 %, and 369 % on average on different sampling days after straw amendment (Fig. S10). Besides elevating the overall DOC concentrations, straw return also increased the fractions of fulvic acid-like, microbial byproduct-like, and humic acid-like (Fig. S11). By analyzing the composition of the soil microbial community, we found that the relative abundance of FeRB (mainly *Geobacteraceae*) increased while methanogens (mainly *Methanocella* and *Methanomicrobia*) decreased (Fig. S12).

Soil MeHg was significantly correlated to k_m and sulfate concentration but not to k_d (Fig. 3a–c). Although a significant relationship was not observed between soil MeHg and dissolved THg for all samples, these two parameters were positively correlated when dissolved THg was lower than 100 ng/L (Fig. 3d). The content of dissolved THg was positively correlated to the content of HCl-extractable Fe(II) ($p < 0.01$; Fig. 3e), and the changes in soil MeHg after straw amendment (%, calculated as the ratio of MeHg difference between RS and CK to MeHg concentration in CK) was also positively correlated to the changes in dissolved THg (ng/L, the difference of THg between RS and CK) (Fig. 3f). Interestingly, soil MeHg, k_d, and dissolved THg were significantly and positively related to the fractions of fulvic acid-like, soluble microbial byproduct-like, and humic acid-like components (except between dissolved THg and fulvic acid-like fractions, Fig. S13a–i).

3.3. Relative contributions of different microbial groups

To identify the roles of key microbial methylators in MeHg production, metabolism inhibition assays were conducted. With SRB inhibited, MeHg concentrations were comparable to those in the CK treatment (Fig. 4a), and sulfate reduction was significantly inhibited (Fig. S14a), indicating the dominant role of SRB in MeHg production in the examined paddy soils. Interestingly, the concentration of HCl-extractable Fe (II) decreased upon the inhibition of sulfate reduction, indicating an interaction between the processes of sulfate reduction and Fe reduction (Fig. S14b). After the inhibition of methanogenesis, methane production decreased (Fig. S14c), while MeHg concentrations were comparable to

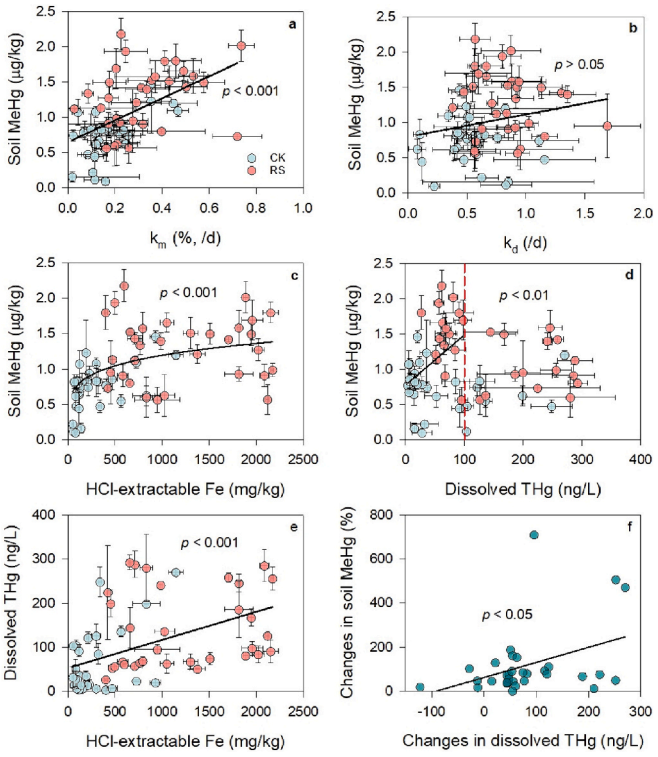


Fig. 3. Temporal variation experiments: Key correlations related to MeHg production, HCl-extractable Fe(II), and dissolved THg. Change in soil MeHg (%) refers to the increasing percentage of MeHg after straw amendment, while the change in dissolved THg is the difference in dissolved THg in RS and CK. For panels a–e, data are presented as mean ± S.D. ($n = 3$); for panel f, since the average was applied to calculate the changes, error bars are not available.

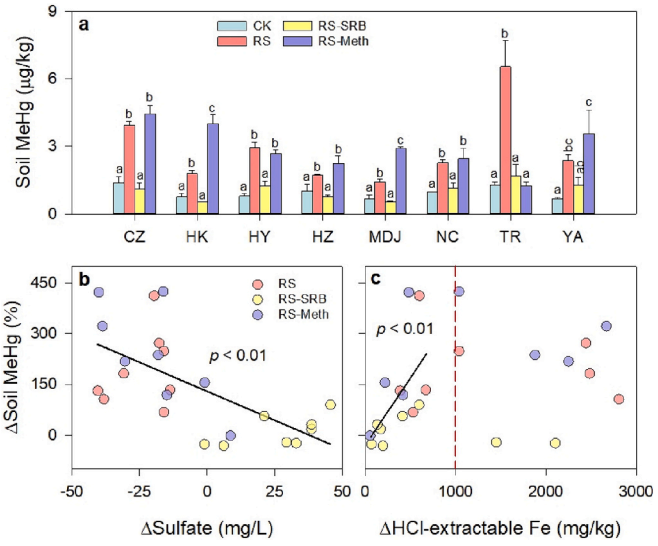


Fig. 4. Metabolism inhibition assays: Concentrations of MeHg (a) and the relationships between changes in soil MeHg (%) and changes in sulfate concentration (b) or HCl-extractable Fe(II) (c). In panel a, data are plotted as mean ± SD. In panels b and c, changes of sulfate or HCl-extractable Fe(II) are the concentration differences between treatments and the control, while the change of soil MeHg is the change ratio of soil MeHg compared with that in the control. Different letters (a–c) indicate significant differences among all treatments ($p < 0.05$).

or higher than those in RS treatment, implying a minor effect of methanogens on net MeHg production or their potential involvement in demethylation (Fig. 4a and Supporting Text 3). Significant relationships were found between changes in soil MeHg (%) and changes in sulfate concentration (Fig. 4b), indicating the dominant role of SRB in the net production of MeHg despite the addition of inhibitors. Although no significant relationship was found between changes in soil MeHg (%) and changes in the contents of HCl-extractable Fe(II) overall, they were significantly correlated when HCl-extractable Fe(II) is lower than 1000 mg/kg (Fig. 4c), suggesting the potential contribution of facilitated Fe reduction to the elevated MeHg production.

4. Discussions

The quantified soil MeHg concentration is a result of Hg methylation and MeHg demethylation, and both processes are related to the biogeochemical cycling of multiple elements such as carbon (C), sulfur (S), iron (Fe), etc. (Tang et al., 2019b; Zhao et al., 2020), which are entangled in complex soil systems. Identifying the limiting factor for net MeHg production in non-Hg-mining impacted paddy soils, therefore, is challenging. Similar to previous studies, we observed significant ($p < 0.05$) increases in soil MeHg production, microbial activities, and Hg mobility upon straw amendment (Sun et al., 2022a; Tang et al., 2019a). By analyzing the changes in key parameters and processes related to Hg methylation in various soils and at different incubation stages with or without straw amendment, we propose that the enhanced activities of SRB play a minor role since the microbial activity or carbon source is not a limiting factor for Hg methylation in non-Hg-mining impacted paddy soils, even though SRB are the primary Hg methylators. In contrast, the supply of Hg is the limiting factor for MeHg production in pristine paddy soils. Therefore, the elevated concentrations of dissolved THg, induced by facilitated Fe reduction and possibly mobilized by straw-derived DOM, were mainly responsible for the increases in soil MeHg after straw amendment, as discussed below.

4.1. Microbial activity is not the limiting factor for Hg methylation

Identifying the primary microbial Hg methylators is the key premise of assessing the relative contribution of microbial activities and IHg supply to MeHg production. The results from the metabolism inhibition assays support that SRB are the primary microbial methylators in the examined rice paddy soils. Specifically, the increases in soil MeHg vanished upon the addition of molybdate that inhibited SRB activities. Additionally, significant negative relationships were observed between changes in soil MeHg and sulfate reduction (Fig. 4a and b). SRB as the predominant Hg methylators in rice paddies has been intensively reported in Hg-mining impacted paddy soils (Li et al., 2019b; Tang et al., 2019a; Wang et al., 2014). This study, for the first time, reveals that SRB also play a dominant role in MeHg production in non-Hg-mining impacted paddy soils on a national scale.

However, it is worth noting that the elevated microbial activities of SRB and other microbial methylators were not mainly responsible for the increases in MeHg production induced by straw amendment. Firstly, the increase in soil MeHg after straw amendment was not correlated to the sulfate concentration or sulfate reduction in the national survey (Fig. S4e and f) or in the metabolism inhibition assays (Fig. 4b, red dots). This suggests that the elevated SRB activities are not the primary driver of the elevated Hg methylation and, consequently, net MeHg production. Secondly, the influence of OM input on *hgcA* gene copies was minor, with a change of <5 %, suggesting that the availability of carbon sources is not a limiting factor for microbial methylators. Prior research has demonstrated significant increases in *hgcA* gene copies following OM input. For instance, Tang et al. reported a 5.6-fold increase (Tang et al., 2019a), while Sun et al. reported several folds (Sun et al., 2022a). The difference in *hgcA* copy changes induced by OM input could be due to the differences in soil microbiome between Hg mining-impacted soils (used in previous studies (Sun et al., 2022a; Tang et al., 2019a)) and non-Hg contaminated soils (used in this study). Specifically, the *hgcA* copies in control soils were approximately 10^8 copies/g soil for bacteria and 10^9 copies/g soil for *Achaea* in this study (Fig. S2), while 10^7 copies/g soil in Sun et al. and 10^6 copies/g soil in Tang et al., suggesting that a much higher abundance of microbial methylators in non-Hg contaminated paddy soils. Thirdly, soil MeHg concentration or k_m (including both Control and RS treatment) was not significantly related to the fractions of Aromatic Protein I and II (data not shown), both of which are known as low molecule DOM and labile carbon sources for microorganisms (Chen et al., 2003). Additionally, other parameters related to microbial activities, such as sulfate reduction and *hgcA* gene copies, exhibited insignificant impacts on MeHg concentrations (Fig. 2 and Table 1). Therefore, due to the relatively high OM contents in rice paddy soils, being 1.3–8.6 % across the 63 studied soils, the supply of carbon source has kept the microbial activities at a high level regardless of straw amendment. The net MeHg production in soils, therefore, responded minorly to changes in the activities of microbial methylators.

4.2. IHg supply is the limiting factor for Hg methylation

With multiple pieces of evidence, we demonstrate that the increased mobility of Hg is a critical contributor to the elevated MeHg concentration. Firstly, the changes in dissolved THg were positively correlated with changes in soil MeHg ($p < 0.05$; Fig. 3f), indicating that elevated Hg mobility is a dominant driver of elevated MeHg production. Previous studies have also demonstrated the mobilization of Hg through detecting Hg species using X-ray absorption near edge structure (XANES) (Tang et al., 2019a) or sequential extraction method (Sun et al., 2022a; Zhu et al., 2015a, 2015b). Secondly, a significant relationship was observed between soil MeHg and dissolved THg in the temporal variation experiment (THg: 3.8–292.7 ng/L; Fig. 3d and Fig. S10) when dissolved THg was lower than 100 ng/L (primarily for CK soils at almost all sampling times; Fig. 3d and Fig. S10), suggesting that the supply of Hg limits Hg methylation in paddy soils without rice straw amendment. Thirdly, results of multiple linear regression and the XGBoost model show that the concentration of HCl-extractable Fe(II) (indicating the reduction of Fe oxides) and pH are the two most important parameters for soil MeHg concentration (Fig. 2b and Table 1), and Fe reduction and pH are recognized as key factors that control Hg mobility (Wang et al., 2021). Unfortunately, the concentrations of dissolved THg or IHg were not available in the spatial variation experiments and, therefore, were not factored into both multiple linear regression and the XGBoost model. A main reason for Hg supply limiting MeHg production lies in the fact that the dissolved THg in pore water is relatively low due to the anaerobic condition. For instance, it ranged from 3.8 to 23.6 ng/L (soil THg: 41–373 μg/kg) in 8 soils on day 7 in this study, and < 50 ng/L (soil THg: 3872 μg/kg; Sun et al., 2022a) and < 50 ng/L (soil THg: 4980 μg/kg; Sun et al., 2022b) in previous studies.

The increases in Hg mobility were primarily induced by the elevated Fe reduction under straw amendment. Our data support the hypothesis that straw amendment facilitates the reduction of Fe(III) (Fig. S3c, Fig. S4c, and Fig. S14b), enhancing the dissolution of Fe oxides and thus the release of the Fe-oxides adsorbed Hg (Wang et al., 2021). This is supported by the significant relationships between dissolved THg and HCl-extractable Fe(II) observed in incubation experiment II (Fig. 3e). Thus, significant relationship between soil MeHg and HCl-extractable Fe (II) could be observed in the spatial variation (incubation experiment I, Fig. S4c), temporal variation (incubation experiment II, Fig. 3c), and metabolism inhibition assays (Fig. 4c). Meanwhile, the significant relationship between dissolved THg and soluble microbial byproduct-like DOM (indicative of microbial activity; Fig. S13h), together with the increased FeRB abundance (Fig. S12), also imply FeRB induced increases in Hg mobility after straw amendment. The roles of *hgcA*-containing methylators and non-putative methylators in Hg methylation have received increasing attention (Liu et al., 2019b; Tang et al., 2020).

While the dominant roles of SRB (Li et al., 2019b; Tang et al., 2019a; Wang et al., 2014) or FeRB (Liu et al., 2018) in MeHg production have been documented, it is still unknown why inconsistent results were observed. Our study suggests that FeRB play a critical yet indirect role in Hg methylation by impacting Hg mobility/availability through the reduction of Fe oxides, though SRB are dominant Hg methylators in paddy soils (as discussed in 4.1). Unfortunately, due to the current technological limitations in regulating Fe reduction using specific inhibitors, we are unable to assess the relative contribution of FeRB to Hg methylation as Hg methylators.

Furthermore, DOM derived from straw decomposition also contributed to increased Hg mobility via an abiotic pathway. Organic matter has been reported to increase Hg mobility (Li et al., 2019a). For instance, our previous study showed that inert HgS decreased from 61.2 to 68.4 % to 41.2–44.7 % during the early stages following straw amendment, while organic complexed Hg rose from <6.5 % to 15.8–21.7 % (Tang et al., 2019a). Here, a significant positive relationship was observed between dissolved THg and humic acid-like DOM (Fig. S13i), suggesting that humus mobilizes Hg in soils. As the straw-derived DOM became more humic at the later stages of incubation (Fig. S11), we observed increases in dissolved THg from day 21 to day 35 (Fig. S10c and d). However, due to technological limitations, it is unable to quantify the bioavailable fractions of IHg among the dissolved THg. Additionally, the Hg present in rice straw may also result in the elevated concentration of THg, as this Hg fraction, i.e., 1.55 ng, accounted for 0.15–0.75 % (the proportion of straw THg to soil THg contents) of the soil THg and averaged 42 % of the contents of dissolved THg in all sampling time in incubation experiment I. By contrast, the contribution of straw-derived MeHg to the increases in soil MeHg should be insignificant, as this MeHg fraction accounted for an average of ~5 % maximum (the proportion of straw MeHg to the increase in soil MeHg) for the increases in soil MeHg. Unfortunately, we are unable to assess the exact contribution of straw-derived Hg to the increases in soil MeHg concentration since it is unknown to what extent Hg in straw was released and methylated.

5. Conclusions

By revealing IHg mobility as the limiting factor for Hg methylation in non-contaminated paddy soils, our study calls for attention to elevated MeHg production induced by Hg mobilization, possibly triggered by intensifying human activities. In paddy soils and other hotspots of Hg methylation, the mobility and bioavailability of Hg are generally low due to the anaerobic conditions that favor the formation of inert HgS and the strong binding between Hg and FeS. Anthropogenic activities, such as fertilization, can mobilize Hg and lead to elevated production of MeHg (Li et al., 2019b). Therefore, Hg mobilization under certain circumstances might be a hidden yet dominant driver for elevated MeHg production.

Additionally, this study also suggests avenues for more effective mitigation of Hg risk. At present, Hg risk mitigation is largely reliant on the Hg emission controls. Nonetheless, decreasing Hg emissions may be less efficacious in reducing MeHg production and accumulation in the rice-paddy systems. This is because Hg emission control primarily focuses on anthropogenic sources, while Hg emissions from natural sources, comparable to the amount of total anthropogenic emissions, are challenging to control. Moreover, annual Hg deposition, $\sim 10^{-4}$ kg/ha, only accounts for a small portion of total Hg in soils ($\sim 10^{-1}$ kg/ka for surface soil). By contrast, Hg immobilization in soils and sediments, for instance, via the application of biochar or other amendments, can directly reduce Hg supply for microbial Hg methylation, serving as a more targeted approach for reducing MeHg production in the specific Hg risk-sensitive systems. The combination of current strategies of Hg emission control and immobilizing Hg, i.e., cutting down Hg supply via deposition and mobilization, may be more effective in decreasing the flux of this potent neurotoxin to our dining tables.

CRediT authorship contribution statement

Huan Zhong: Writing – review & editing, Supervision, Project administration, Conceptualization. **Yao Su:** Investigation, Formal analysis, Data curation. **Xinda Wu:** Writing – review & editing, Methodology, Data curation. **Luís Nunes:** Writing – review & editing, Validation, Formal analysis. **Chengjun Li:** Writing – review & editing. **Yunyun Hao:** Data curation. **Yu-Rong Liu:** Resources, Methodology. **Wenli Tang:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors have no competing interests to declare.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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